

Quayle (2006) — One-Page Formula & Concept Sheet

Print target: one page (briefing / one_page CSS)

Purpose: Fast-reference sheet for the Quayle assortative-growth mechanism and the distinctions that matter for VECTOR.

The Core Causal Chain

$$\text{property} \rightarrow \text{similarity } x_{ij} \rightarrow \text{homophily } \alpha \rightarrow \text{attachment } \Pi \rightarrow \text{assortativity } r \rightarrow \text{community structure } Q$$

Do not conflate these objects. Similarity x_{ij} is a **pairwise property**; homophily α is a **generative preference strength**; assortativity r is a **network-level realized outcome**. In particular:

$$x_{ij} \neq \alpha \neq r$$

Two stochastic networks generated with the same α need not realize exactly the same r .

Governing Attachment Rule — Equation (4)

$$\Pi(k_j, x_{ij}) = \frac{f(k_j)e^{-\alpha(1-x_{ij})}}{\sum_l f(k_l)e^{-\alpha(1-x_{il})}}.$$

Read this as:

$$\text{structural attractiveness} \times \text{similarity attraction} \rightarrow \text{normalized attachment probability}$$

Here i is the incoming vertex, j is the candidate vertex, l indexes all eligible candidates, k_j is candidate j 's degree, $f(k_j)$ is its structural attractiveness, and Π is the probability that i attaches to j . Standard BA is the special implementation $f(k_j) = k_j$.

The similarity weight is $e^{-\alpha(1-x_{ij})}$. At $\alpha = 0$, similarity has no effect. Larger α increasingly penalizes dissimilar candidates.

Similarity Functions

Discrete properties: $x_{ij} = 1$ for the same property and $x_{ij} = 0$ for different properties. Therefore:

$$w_{\text{same}} = 1, \quad w_{\text{different}} = e^{-\alpha}.$$

Hierarchical properties — Eq. (12):

$$x_{ij} = 1 - \frac{g_{ij}}{G}.$$

g_{ij} measures hierarchical separation and G is the maximum hierarchy depth. Closer common ancestry means greater similarity.

Scalar / continuous properties — Eq. (15):

$$x_{ij} = 1 - |p_i - p_j|$$

for appropriately scaled p . Substituting into Eq. (4) gives Eq. (16):

$$\Pi(k_j, p_i, p_j) = \frac{f(k_j)e^{-\alpha|p_i - p_j|}}{\sum_l f(k_l)e^{-\alpha|p_i - p_l|}}$$

This scalar form is especially portable: **attribute distance** $\rightarrow$ **exponential similarity preference** $\rightarrow$ **attachment probability**.

Homophily α vs. Assortativity r

For the symmetric discrete-property ensemble, Quayle derives:

$$r = \frac{1 - e^{-\alpha}}{1 + (n_p - 1)e^{-\alpha}} \quad (\text{Eq. 11})$$

where n_p is the number of discrete properties. Thus $\alpha \rightarrow r$: microscopic preference generates macroscopic assortment.

The inverse is:

$$\alpha = \ln \left(\frac{1 + (n_p - 1)r}{1 - r} \right) \quad (\text{Eq. 28})$$

But this **does not define α as assortativity**. It infers a candidate α from observed r **only under Quayle’s discrete symmetric ensemble assumptions**.

Assortativity vs. Community Structure

Assortativity r measures property mixing across edges. Modularity Q measures topological community structure:

$$Q = \sum_u (e_{uu} - a_u^2).$$

Quayle’s simulations show that r rises before strong Q : **assortative mixing can exist before sharply separated communities emerge**. Hierarchical similarity can generate hierarchical community structure.

What to Remember for VECTOR

- **Similarity x_{ij}** : how alike two candidate entities are.
 - **Homophily α** : how strongly similarity changes attachment odds.
 - **Attachment II**: the stochastic local choice rule.
 - **Assortativity r** : realized network-level mixing after choices accumulate.
 - **Modularity Q** : downstream community structure.
 - The similarity metric is a **modeling choice**, not a natural law.
 - Quayle uses **one-mode growth, degree attraction, and no hard capacity**.
 - Capacity constraints, bipartite affiliation, or reassignment change the formation process.
 - Therefore **do not transplant Quayle’s Eq. (11) or Eq. (28) into VECTOR without re-deriving or empirically validating the new $\rho \rightarrow$ realized-assortment relationship**.
- Five-year memory:** Quayle turns “similar things tend to connect” from a descriptive observation into a generative probability rule, then follows the consequences upward from local attachment to assortative mixing to community structure.
- Source:** Quayle, A. P., Siddiqui, A. S., & Jones, S. J. M. (2006). *Modeling network growth with assortative mixing*. European Physical Journal B, 50, 617–630.